

The low-temperature-sensitive (LOT) genes are a group of genes in the yeast *Saccharomyces cerevisiae* that are expressed at higher levels as temperature decreases. These genes perform various functions within the cell, ranging from glucose metabolism to ribosome stabilization to RNA processing.

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To characterize the kinetics of LOT6p, two experiments were conducted. In the first experiment, LOT6p was mixed with varying amounts of cofactor and a constant amount of the substrate 2,6-dichlorophenolindophenol (DCPIP), and the initial reaction rate for each cofactor concentration was plotted. In the second experiment, cofactor concentration was held constant while the DCPIP concentration was varied, and the reaction rate as a function of DCPIP concentration was recorded. Neither experiment showed any evidence of cooperativity.

A third experiment showed that the molecule resveratrol competitively inhibited LOT6p with respect to both substrates and that as the concentration of resveratrol was increased, the affinity of LOT6p for resveratrol decreased.

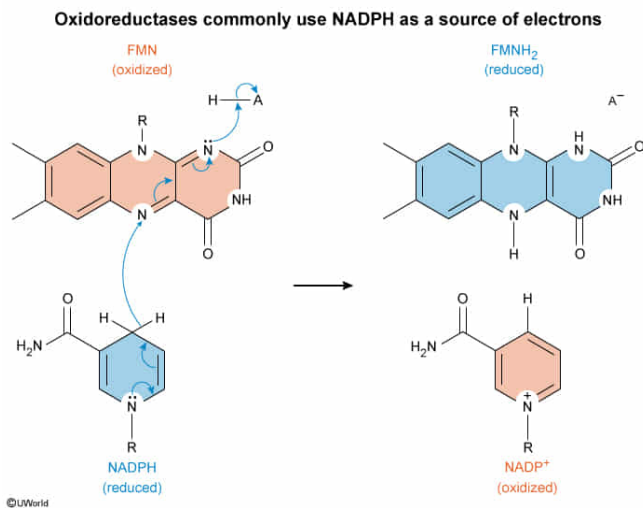
Which of the following molecules would best serve as the cofactor that reduces FMN in the LOT6p mechanism described in the passage?

- ☐ A. Ubiquinone [15%]
- ☐ B. Acetyl-CoA [4%]
- ☐ C. Mg²⁺ [19%]
- ☒ D. NADPH [61%]

Correct

60% answered correctly

Explanation:



Oxidoreductases are enzymes that catalyze **oxidation-reduction** (redox) reactions, in which **electrons are transferred** from one molecule to another. **Cofactors** are non-amino acid molecules that are required for a protein to function. Some cofactors can be further classified as **cosubstrates**: molecules that directly participate in an enzyme-catalyzed reaction in addition to the substrate of interest.

According to the passage, **LOT6p is an oxidoreductase** enzyme that reduces quinones (and similar molecules) by taking electrons from a cofactor and ultimately giving those electrons to the quinone (or alternate substrate). The cofactor that loses electrons passes those electrons to FMN, where the electrons are held temporarily before they are eventually passed to the final molecule. **Oxidoreductases commonly use the cofactors NADH, NADPH, or FADH₂** as a **source** of electrons. Of the available choices, only NADPH could be used as a source of electrons to reduce quinones.

(Choice A) **Ubiquinone** (also called coenzyme Q, CoQ, or UQ) is a cofactor used in the **electron transport chain**. Ubiquinone is the fully oxidized form of the molecule and cannot readily donate electrons to FMN. Instead, ubiquinone is likely to **receive electrons** from FMNH₂ to become ubiquinol (also called UQH₂ or QH₂).

(Choice B) Acetyl-CoA is a coenzyme involved in many **metabolic reactions**, but it is not generally used as an electron source in redox reactions.

(Choice C) Mg^{2+} is a cofactor found in the active sites of many enzymes, but it is fully oxidized and cannot transfer electrons to other molecules.

Educational objective:

Oxidoreductases commonly use NADH, NADPH, or FADH_2 as sources of electrons for transfer to other molecules.

Time spent:0 sec

Subject:
Biochemistry

QID:403712

System:
Enzymes

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A third experiment showed that the molecule resveratrol competitively inhibited LOT6p with respect to both substrates and that as the concentration of resveratrol was increased, the affinity of LOT6p for resveratrol decreased.

The mechanism by which LOT6p reduces quinones can be best described as:

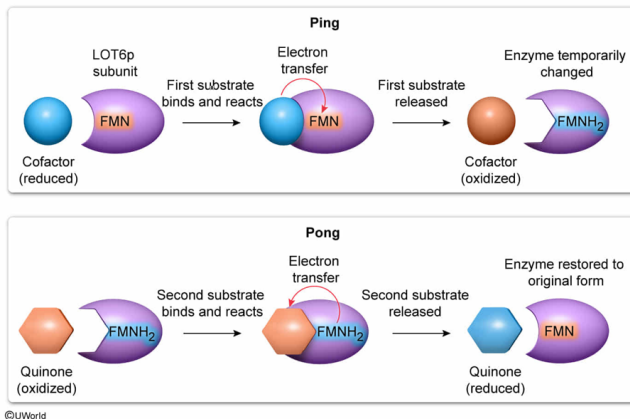
- ☐ A. a random ping-pong mechanism. [3%]
- ✓ ☒ B. an ordered ping-pong mechanism. [57%]
- ☐ C. random formation and dissociation of a ternary complex. [6%]
- ☐ D. ordered formation and dissociation of a ternary complex. [32%]

Correct

57% answered correctly

Explanation:

Ping-pong enzyme mechanisms



Many enzymatic reactions involve **more than one substrate**. Two-substrate reactions can proceed through a variety of mechanisms. Two of the most common mechanisms are:

- Formation of a **ternary complex**, in which **three** molecules are bound together (as opposed to a binary complex involving only two molecules). In the case of two-substrate reactions, the three molecules are the enzyme and the two substrates. Once all three molecules are bound, catalysis proceeds and the resulting products are released. **Some ternary complexes form in a random order** (ie, the order in which substrates bind to the enzyme does not matter). **Other ternary complexes must form in a specific order**, with one substrate always binding the enzyme first and then the other substrate binding second.
- **Ping-pong mechanisms**, in which **one substrate binds** the enzyme first and **reacts, changing the structure of the enzyme itself**. This substrate then leaves the active site, allowing the **second substrate to enter and react, restoring the enzyme to its original form**. Ping-pong mechanisms are **always ordered** (ie, one substrate must bind first, and then the other).

The passage describes a mechanism in which one substrate (ie, the cofactor) enters the active site and reduces FMN to FMNH₂. The cofactor then leaves, allowing the quinone substrate to enter the active site, where the quinone is reduced and FMNH₂ reverts to FMN. **This is an example of an ordered ping-pong mechanism.**

(Choice A) Ping-pong mechanisms *cannot occur in a random order*. One particular substrate must always bind, react, and leave before the other substrate can bind to the enzyme.

(Choices C and D) In the described mechanism, the first substrate (ie, the cofactor) *leaves the active site* before the second substrate enters. Therefore, no ternary (ie, three-membered) complex forms in this mechanism. (Note that as a prosthetic group, FMN is part of the LOT6p enzyme and *does not* constitute a third molecule in any of the complexes formed.)

Educational objective:

Enzymatic reactions with two substrates typically proceed either by formation of a ternary complex (which may occur in a random or a specific order) or by a ping-pong mechanism, in which one substrate binds, reacts, and leaves before the other substrate binds.

Time spent:0 sec
Subject:
Biochemistry

QID:403713
System:
Enzymes

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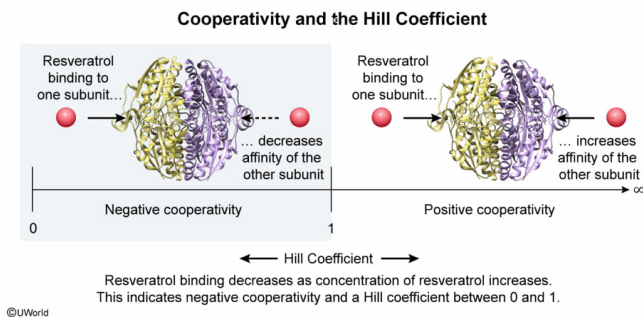
Based on the information in the passage, the Hill coefficient for LOT6p binding to resveratrol is:

- ☐ A. less than 0. [19%]
- ✓ ☒ B. between 0 and 1. [47%]
- ☐ C. exactly equal to 1. [23%]
- ☐ D. greater than 1. [9%]

Correct

47% answered correctly

Explanation:



Some proteins have multiple **subunits** that perform the same function (eg, homodimers). Because they have multiple binding sites, these proteins can exhibit **cooperativity**, in which ligand **binding to one subunit alters the affinity of other subunits for the ligand**. The extent of cooperativity is measured by the **Hill coefficient**. A Hill coefficient **greater than 1 indicates positive cooperativity**, in which binding of one subunit increases the affinity of other subunits for the ligand. A Hill coefficient **between 0 and 1 indicates negative cooperativity**, in which binding to one subunit decreases the affinity of other subunits for the ligand. A Hill coefficient of exactly 1 indicates no cooperativity.

The passage states that as the concentration of resveratrol increased, the **affinity** of LOT6p (a homodimer) **for resveratrol decreased**. This indicates that resveratrol binding to one subunit of LOT6p caused a decrease in the affinity of the other subunit for resveratrol. Therefore, the LOT6p-resveratrol interaction exhibits **negative cooperativity**, and the **Hill coefficient for this interaction is between 0 and 1**.

(Choice A) Despite the term "negative cooperativity" for interactions in which one binding event weakens the affinity of other subunits, Hill coefficients always have positive values (ie, greater than 0). Negative cooperativity is indicated by a Hill coefficient between 0 and 1.

(Choice C) A Hill coefficient of exactly 1 would indicate no cooperativity, in which case increasing resveratrol concentration would have no effect on the affinity of LOT6p for resveratrol.

(Choice D) A Hill coefficient greater than 1 would indicate positive cooperativity, in which increasing resveratrol concentration would increase the affinity of LOT6p for resveratrol.

Educational objective:

Cooperativity refers to changes in the affinity of one subunit of a protein for its ligand when other subunits are bound. Positive cooperativity occurs when binding one subunit increases the affinity of other subunits, indicated by a Hill coefficient greater than 1. Negative cooperativity occurs when binding to one subunit decreases the affinity of other subunits, indicated by a Hill coefficient between 0 and 1.

Time spent:0 sec**Subject:**
Biochemistry**QID:**403714**System:**
Enzymes

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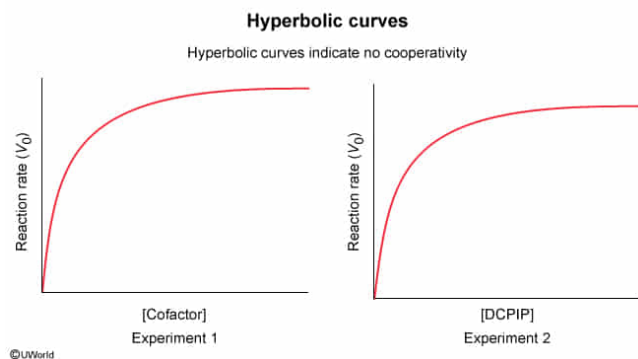
What were the most likely shapes of the curves generated by the first two experiments described in the passage?

- ✔ ☒ A. Both experiments generated hyperbolic curves [76%]
- ☐ B. The first experiment generated a hyperbolic curve and the second generated a sigmoidal curve [8%]
- ☐ C. The first experiment generated a sigmoidal curve and the second generated a hyperbolic curve [7%]
- ☐ D. Both experiments generated sigmoidal curves [8%]

Correct

77% answered correctly

Explanation:



Many enzymes follow **Michaelis-Menten kinetics**, in which the rate of the reaction depends on the concentration of the substrate, and a plot of rate as a function of substrate concentration yields a **hyperbolic curve**. However, enzymes that exhibit **positive cooperativity** yield **sigmoidal** ("S"-shaped) curves that reflect how affinity increases as substrate concentration increases.

Because LOT6p is a **homodimer**, it is **theoretically capable of exhibiting cooperativity**. However, the passage states that **neither of the first two experiments performed showed any evidence of cooperativity**. These experiments varied the concentration of one LOT6p substrate while holding the other constant, and reaction rate was plotted as a function of the concentration of the substrate that was varied. Because no cooperativity was exhibited, it is likely that **LOT6p follows Michaelis-Menten kinetics** with respect to both substrates. Therefore, it is most likely that both experiments generated **hyperbolic curves**.

(Choices B, C, and D) If either of the graphs were sigmoidal, this would be evidence of positive cooperativity in at least one of the experiments.

Educational objective:

Enzymes that follow Michaelis-Menten kinetics generate hyperbolic curves when the reaction rate is plotted as a function of substrate concentration. Enzymes that exhibit positive cooperativity generate sigmoidal curves.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403715

System:
Enzymes

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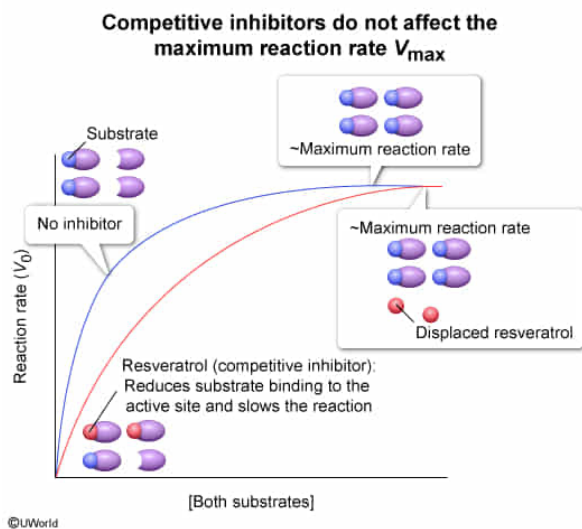
Which of the following effects on the maximum possible reaction rate is expected from the third experiment performed?

- ☐ A. The maximum reaction rate increases [4%]
- ☐ B. The maximum reaction rate decreases [28%]
- ☒ C. The maximum reaction rate does not change [64%]
- ☐ D. More information is needed to determine the effect on the maximum reaction rate [2%]

Correct

65% answered correctly

Explanation:



Different enzyme inhibitors work by different mechanisms and, as such, have different effects on the rate of the enzyme-catalyzed reaction. **Competitive inhibitors** bind to free enzymes and thereby **prevent substrates from binding** to the active site. This interaction decreases the reaction rate of the enzyme under **nonsaturating conditions** because fewer substrate molecules are bound to an active site.

However, when the concentration of substrate is sufficiently high, it can outcompete the inhibitor and saturate the enzyme (ie, cause all active sites to be bound by substrate). Consequently, when a **competitive inhibitor** is added to a system containing enzyme and substrate, the substrate concentration required to approach the maximum possible reaction rate $V_{\max}$ increases but $V_{\max}$ **itself does not change**.

According to the passage, the third experiment involved the competitive inhibitor resveratrol. As a competitive inhibitor, **resveratrol has no effect on the maximum rate of the reaction** catalyzed by LOT6p.

(Choice A) If resveratrol caused an increase in the maximum reaction rate, it would be an activator instead of an inhibitor.

(Choice B) Inhibitors cause a decrease in the maximum reaction rate only when they are not competitive inhibitors (ie, **uncompetitive** and **mixed inhibitors**, including **noncompetitive** inhibitors). Resveratrol is a competitive inhibitor.

(Choice D) The information stating that resveratrol competitively inhibits LOT6p is sufficient to determine that it has no effect on the maximum reaction rate.

Educational objective:

Competitive inhibitors bind to free enzymes and prohibit substrate binding to the active site. Because sufficiently high concentrations of substrates can outcompete the inhibitor, competitive inhibitors change the amount of substrate needed to approach the maximum reaction rate $V_{\max}$ but do not alter $V_{\max}$ itself.